Supply Chain Control Tower

Executive Summary

1. Project Overview

This project analyzes 180,519 orders from the DataCo Smart Supply Chain dataset (2015-2018) to identify logistics bottlenecks, score supplier reliability, optimize inventory, and predict late delivery risk. The analysis spans 6 Python notebooks, 5 SQL scripts, and an interactive dashboard.

2. Key Metrics

Total Orders	**180,519**
On-Time Delivery Rate	**42.7%**
Late Delivery Rate	**57.3%**
Average Lead Time (Actual)	**3.5 days**
Average Shipping Delay	**0.57 days**
Total Revenue	**$36.8M**
Total Profit	**$5.9M**
Revenue Lost to Canceled/Fraud	**$1.57M**
Unique Products	**118**
Unique Customers	**20,652**
Estimated Inventory Turnover	**12.2x/year**

3. Key Findings

H1 - Shipping Mode vs Delays

Rejected. Standard Class has 60.2% on-time rate vs First Class 0.0% (data anomaly).

H2 - Geographic Variance

Partially confirmed. Central Africa worst (60.7% late), Canada best (51.9% on-time).

H3 - Pareto on Suppliers

Confirmed. Top 20% of origin locations cause 71% of all late orders.

H4 - Late Delivery Margins

Not confirmed. On-time profit $33.08 vs Late $32.67 (negligible difference).

H5 - Seasonal Peaks

Not confirmed. Q1 highest ($9.47M), Q4 lowest ($8.47M) -- no holiday peak.

ML Model

Near-random (ROC-AUC 0.525). Static features insufficient to predict future delays.

4. Recommendations

1. Prioritize Standard Class Shipping

Standard Class has the best on-time performance (60.2%). First Class and Same Day should be reviewed for process issues or removed as options.

2. Target Geographic Hotspots

Central Africa, Southeast Asia, and South Asia have late rates above 58%. Consider regional distribution hubs or alternative logistics partners.

3. Focus on Top 20% of Origins

71% of late orders come from just 20% of origin locations. Audit and improve processes at these locations for maximum impact.

4. Improve Data Quality for ML

The predictive model failed because static order features don't capture time-varying factors. Integrate real-time logistics data (weather, port congestion, carrier status) for production-grade prediction.

5. Monitor ABC-XYZ Inventory Segmentation

6 products (A-items) drive 70% of revenue. These should have tight inventory control. 115 products are X-stable (predictable demand) -- optimize reorder points.

6. Address $1.57M Revenue Loss

Canceled and suspected fraud orders represent $1.57M in lost revenue. Investigate root causes in processing and payment workflows.

7. Implement Supplier Scoring Dashboard

Composite scoring (on-time rate, consistency, product breadth) enables data-driven supplier decisions. Phase out lowest-scoring origins.